

Timeliness of progeny-testing through AI and percentage of bulls returned to service (abstract 1020)

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Successful progeny testing (PT)

- **Select bulls with high pedigree merit**
- **Early semen collection**
- **Rapid semen distribution and use in milk-recorded herds**
- **Daughters with unique ID, recorded pedigrees, and kept until milking**

PT accuracy and impact

- **Accuracy of PT evaluations**
 - Number of daughters
 - Daughter distribution across herds
- **Impact on genetic improvement**
 - Number of bulls sampled
 - Intensity of culling of PT bulls

Objectives

- **Summarize characteristics of PT programs related to timeliness of genetic evaluation**
- **Examine distribution of PT daughters**
- **Determine selection intensity after evaluation of PT bulls**

Data sources

- **National database at AIPL, ARS, USDA**
 - Yield information from DHI
 - ID information from DHI and breed associations
- **Bull entry into PT programs and semen distribution dates from NAAB**

PT definition

- **AI organization in US**
 - (exception: Semex USA)
- **Bulls aged 10-29 mo at semen release**
- **NAAB sampling controller codes 1-222**
- **Daughters born 8-42 mo after bull entry into AI sampling**
 - Usable 1st-lactation record
 - First 125 based on calving date

Number of PT bulls (major studs)

Year	Ayr	BS	Gue	Hol	Jer	MS
1970	2	2	19	233	17	1
1975	3	8	20	563	35	2
1980	13	33	23	826	23	1
1985	19	16	41	969	52	1
1990	7	30	40	1191	103	3
1995	14	30	26	1373	134	3
1998	8	21	18	1219	95	2

Bull age (mo) at semen distribution

Year*	Ayr	BS	Gue	Hol	Jer
1975-79	19	16	17	16	17
1980-84	19	16	16	16	17
1985-89	19	15	16	16	16
1990-94	18	16	16	16	16
1995-99	18	16	16	16	16

*Year of bull entry into AI sampling

Bull age SD (mo) at semen distribution

Year*	Ayr	BS	Gue	Hol	Jer
1975-79	3.6	3.8	3.6	2.8	3.2
1980-84	3.7	3.1	2.9	3.0	3.3
1985-89	4.9	2.5	3.2	2.8	2.9
1990-94	4.1	3.3	3.4	2.4	3.4
1995-99	3.2	2.8	2.4	2.4	2.8

*Year of bull entry into AI sampling

Bull age (mo) at daughter birth

Year*	Ayr	BS	Gue	Hol	Jer
1970-74	39	42	38	33	38
1975-79	39	40	37	32	38
1980-84	39	40	35	32	35
1985-89	39	38	35	30	33
1990-94	36	36	34	29	32

*Year of bull entry into AI sampling

Bull age SD (mo) at daughter birth

Year*	Ayr	BS	Gue	Hol	Jer
1970-74	6.4	5.0	4.6	5.7	4.6
1975-79	4.8	3.8	4.1	5.1	4.9
1980-84	4.2	3.8	3.8	5.2	4.4
1985-89	4.3	4.0	4.4	4.5	3.5
1990-94	4.0	4.3	3.9	3.2	3.9

*Year of bull entry into AI sampling

Bull age (mo) at daughter calving

Year*	Ayr	BS	Gue	Hol	Jer
1970-74	68	70	66	61	64
1975-79	69	68	65	60	64
1980-84	68	68	63	60	62
1985-89	68	66	62	57	59
1990-94	65	64	62	56	58

*Year of bull entry into AI sampling

Bull age SD (mo) at daughter calving

Year*	Ayr	BS	Gue	Hol	Jer
1970-74	6.2	5.5	4.7	5.9	4.9
1975-79	4.7	3.8	4.1	5.3	5.1
1980-84	4.3	4.0	4.0	5.4	4.4
1985-89	4.2	4.0	4.5	4.6	3.7
1990-94	4.1	4.4	4.0	3.3	4.1

*Year of bull entry into AI sampling

PT daughter age (mo) at calving

Year*	Ayr	BS	Gue	Hol	Jer
1970-74	29	29	28	28	27
1975-79	30	28	28	28	26
1980-84	29	29	28	28	26
1985-89	29	28	28	27	26
1990-94	28	28	27	26	26

*Year of bull entry into AI sampling

PT daughters (no.)

Year*	Ayr	BS	Gue	Hol	Jer
1960-64	57	51	48	67	56
1965-69	100	56	64	55	60
1970-74	72	71	53	47	48
1975-79	93	45	52	48	61
1980-84	61	30	52	48	64
1985-89	44	43	40	56	60
1990-94	44	28	31	61	49

*Year of bull entry into AI sampling

Herd distribution (%) by herd PT use

Year*	Herd PT use				
	0%	1-19%	20-49%	50-99%	100%
1965	85	11	4	1	<.1
1975	68	22	9	1	<.1
1985	56	32	11	1	.1
1990	51	35	12	2	.1
1995	41	39	16	3	.3
1998	39	38	18	5	.4

*Year of first calving

PT cows (%) by herd PT use

Year*	Herd PT use				
	<10%	10-19%	20-49%	50-99%	100%
1965	18	32	38	11	.1
1975	19	31	44	6	.1
1985	20	33	40	7	.3
1990	18	32	38	12	.6
1995	14	27	42	15	1.0
1998	11	24	41	23	1.1

*Year of first calving

PT bulls (%) returned to AI service

Year*	Ayr	BS	Gue	Hol	Jer
1963-67	71	48	36	44	46
1968-72	57	88	44	45	50
1973-77	67	62	32	39	56
1978-82	46	51	36	20	38
1983-87	37	46	28	19	26
1988-92	23	19	22	14	19

*Year of bull entry into AI sampling

Conclusions

- **Slight increase in age of PT bull at semen release**
- **Decrease in age of PT bulls at daughter birth**
 - 4.5 mo for Holsteins and Jerseys
- **Decrease in age of PT bulls at daughter calving**
 - 6 mo for Holsteins
 - 5.5 mo for Jerseys

Conclusions (cont.)

- Decrease in herds with no PT records usable for genetic evaluation
 - 85% → 39%
- Increase in herds with usable PT records
 - 1-19% PT use: 11% → 38%
 - >50% PT use: 1% → 5%
- 1 in 7 PT bulls returned to active service
not 1 in 10

